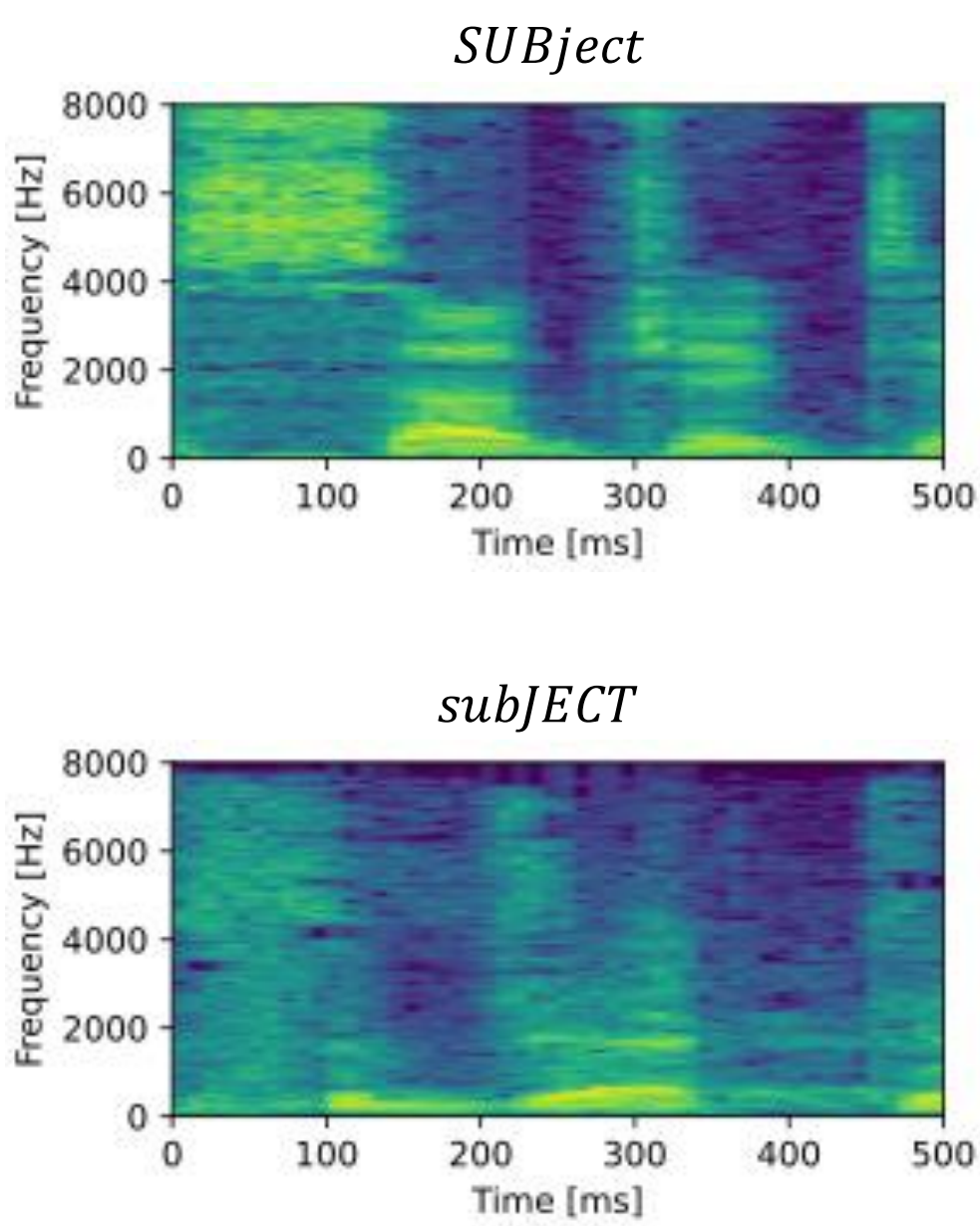


How Does a Deep Neural Network Look at Lexical Stress?

Itai Allouche, Itay Asael, Rotem Rousso, Vered Dassa, Ann Bradlow, Seung-Eun Kim, Matthew Goldrick, Joseph Keshet

Introduction

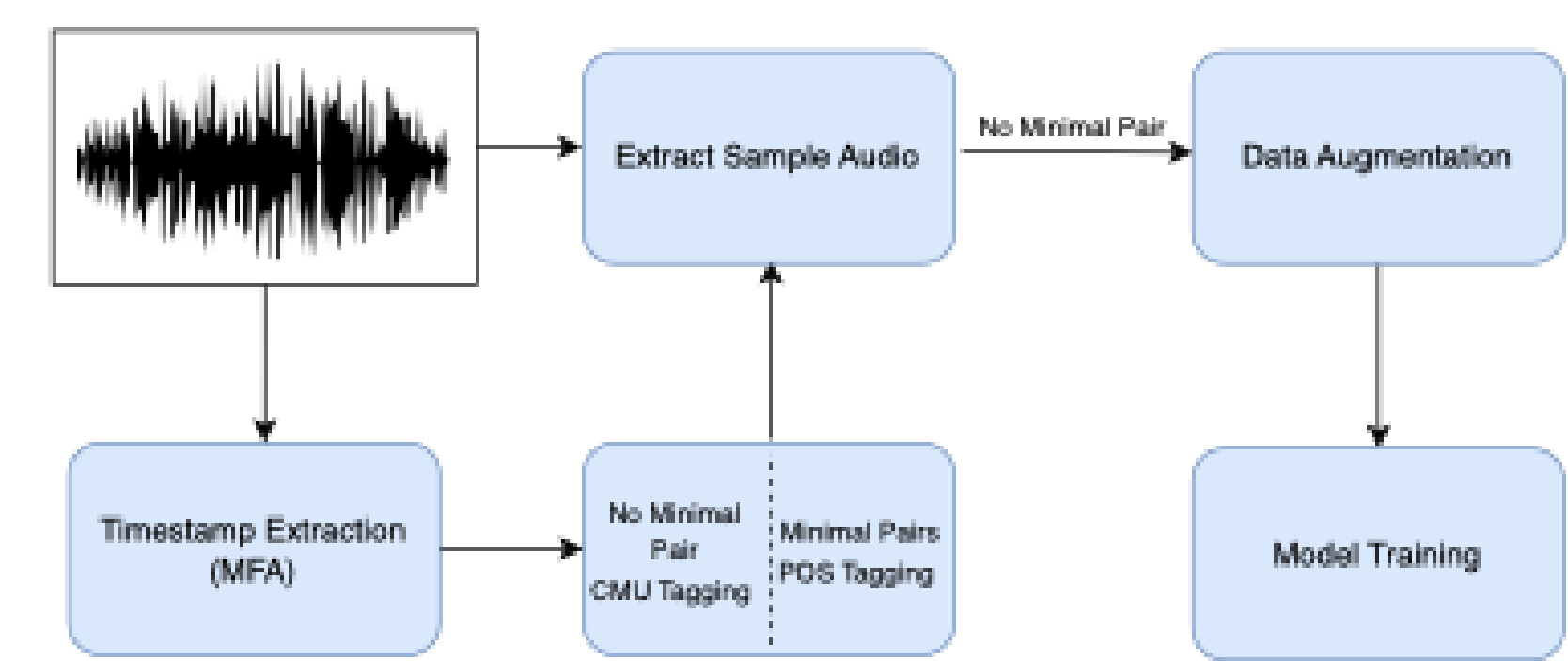
- Neural networks have achieved significant success in speech processing.
- Despite this success, they often operate as black boxes.
- The work examines these issues specifically in the context of lexical stress - emphasis on a syllable that can change word meaning (e.g., "*PRE*sent" vs. "*pre*SENT"), crucial in languages like English and Hebrew
- Minimal pairs are word pairs with similar phonetic structures but different primary stress placements
- Stress differentiation is key to distinguishing between them.



Contributions

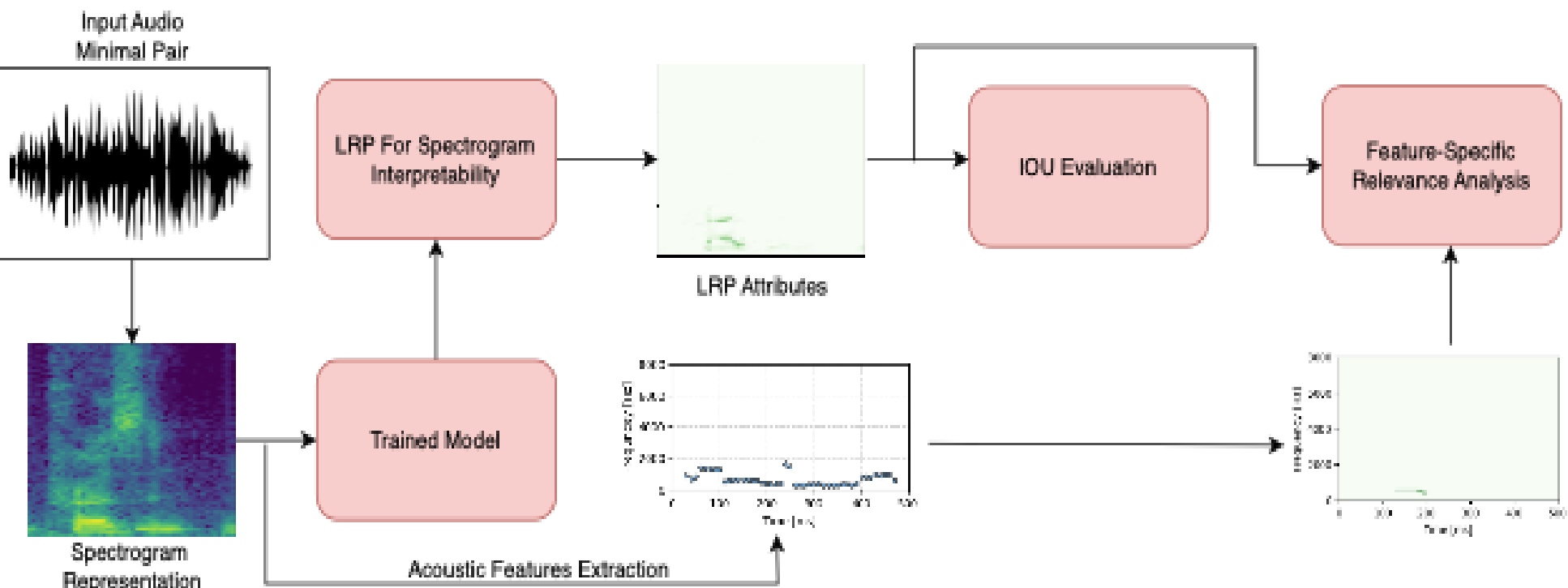
- Created a fully automatic dataset of disyllabic words (minimal and non-minimal pairs) from large speech corpora with augmentation.
- Developed a state-of-the-art CNN classifier for lexical stress, achieving 92% accuracy and cross-linguistic generalization.
- Applied Layerwise Relevance Propagation (LRP) to analyze the spectral features these classifiers use to achieve high accuracy.

Data Preparation and Training



- Minimal and no minimal pairs were extracted from serval corpus using Montreal Forced Aligner (MFA)
- Minimal pairs were tagged with part-of -speech labels using the SpaCy package
- Data augmentation techniques were applied to expand the dataset
- The model was trained on no minimal pairs and evaluated on both minimal and no minimal pairs

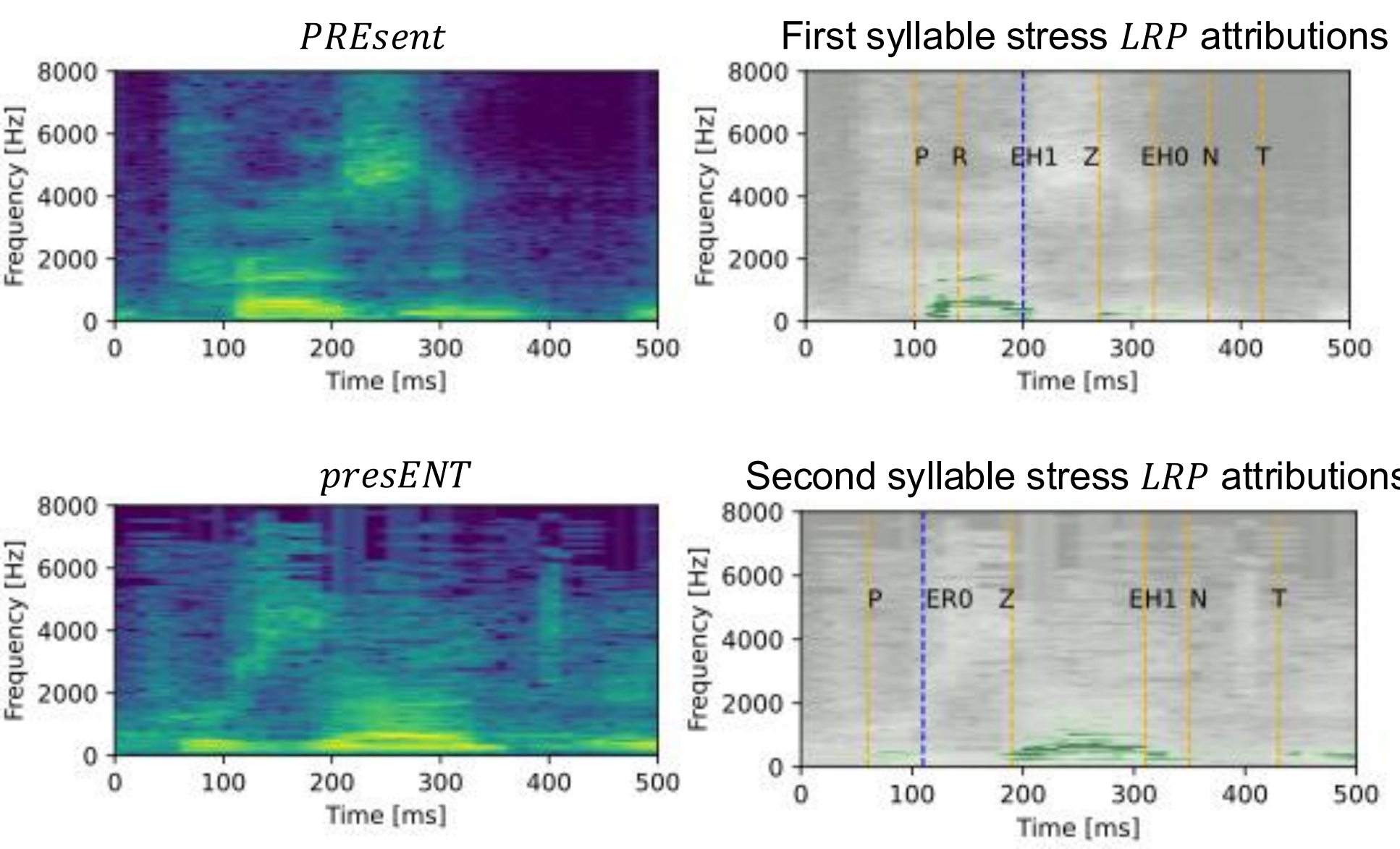
Model Interpretability and Acoustic Feature Relevance



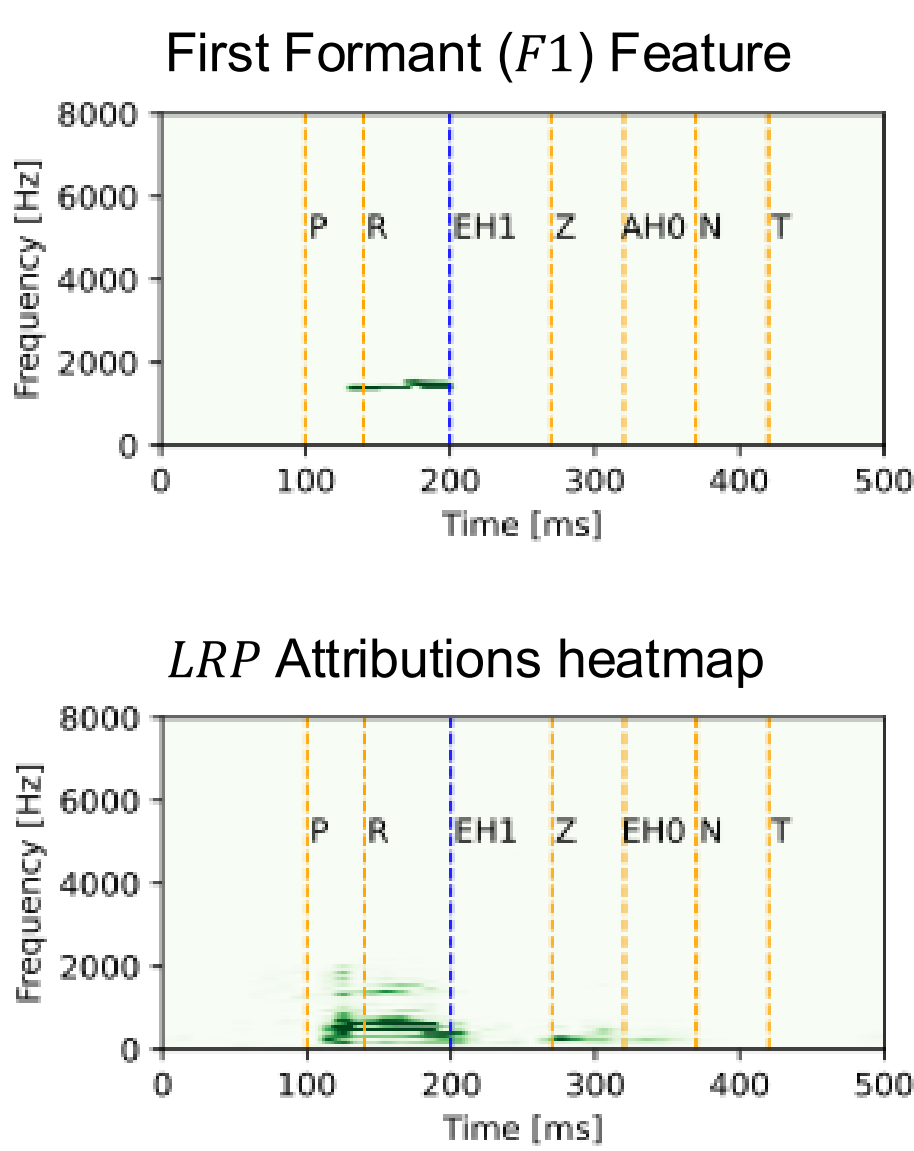
- Layer-wise Relevance Propagation (LRP) method was applied to highlight important key regions for classification
- Pitch (F_0) and the first three formants (F_1, F_2, F_3) were extracted using Praat Software
- Evaluation metrics were applied to LRP results and acoustic features

Evaluation

- LRP heatmaps
 - Bounding boxes were defined around the most prominent syllables.
 - Intersection Over Union (IOU) was used to assess overlap between bounding boxes and LRP heatmap



- Acoustic features relevance
 - Introduced a new interpretability measure alongside the IOU metric.
 - Assesses the model's focus on key acoustic features.
 - Pearson Correlation Coefficient (PCC) r was used to assess the similarity between acoustic features and the LRP heatmap.
 - The p value was used to assess the statistical significance of the correlation.



Results

- Comparing model performance across different languages and datasets
- Test set discrepancy likely due to SpaCy's tagging accuracy variations

Test Set	Classifier	Accuracy (%)
Minimal pair	LeNet	85
	VGG11	86
	VGG16	87
	VGG19	87
	ResNet18	86
No minimal pair	LeNet	90
	VGG11	91
	VGG16	92
	VGG19	92
	ResNet18	90

Test Set	Classifier	Accuracy (%)
Dutch	VGG16	81
	ResNet18	79
	VGG16 after fine-tuning	93
	ResNet18 after fine-tuning	93
Hebrew	VGG16	68
	ResNet18	73
	VGG16 after fine-tuning	88
	ResNet18 after fine-tuning	85

- IOU score (μ) across different LRP methods

Classifier	LRP Method	μ_{vowel}		$\mu_{\text{not vowel}}$	
		Stress	No stress	Stress	No stress
VGG16	LRP_{CMP}	0.432	0.135	0.276	0.156
	LRP_{α_1}	0.419	0.129	0.279	0.173
	LRP_{ϵ}	0.428	0.137	0.27	0.164
	LRP_{z}	0.301	0.154	0.318	0.226
ResNet18	LRP_{CMP}	0.37	0.136	0.298	0.195
	LRP_{α_1}	0.361	0.128	0.255	0.255
	LRP_{ϵ}	0.34	0.13	0.275	0.254
	LRP_{z}	0.326	0.12	0.204	0.364

- r and p values for heatmaps based on different features in the stressed vowel region

Method	Acoustic Feature	Mean r	Mean p
VGG16 + LRP_{CMP}	F_1	0.366	0.003
	F_1, F_2	0.359	0.003
	F_1, F_2, F_3	0.305	0.002
	F_1, F_3	0.277	0.001
	F_0, F_1, F_2	0.245	0.001
	F_0, F_1, F_2, F_3	0.237	0.001
	F_0, F_1, F_3	0.205	0.001
	F_0, F_2	0.154	0.003
	F_2, F_3	0.136	0.008
	F_2	0.154	0.008
	F_0	0.116	0.004
	F_0, F_3	0.114	0.003
	F_3	0.048	0.02

- Relative distribution of pixels in the stressed vowel region of the residual *LRP_{cmp}* heatmap for VGG16

Region	Initial stressed Vowel	Final stressed Vowel
Between F_0, F_1	0.93	0.886
Between F_1, F_2	0.07	0.1134
Between F_2, F_3	0	0
Above F_3	0	0

